

General guidelines

Data files supplied by your instructor

- The data files needed for each case study are identified in the specifications for each study, and your instructor should make these starting files available to you.

Notebook created by you

- For each case study, create one Jupyter Notebook.
- When creating the filename for your Notebook, please use the convention specified by your instructor. Otherwise, name the Notebook *first_last_case_study* where *first_last* specifies your first and last name and *case_study* specifies the name of the case study.
- Use Code cells to store the code for your analysis.
- Use Markdown cells to create headings that organize your Notebook and to create headings that provide short answers to questions.
- Use Raw cells to store longer sections of raw text.
- Use comments or raw text to describe your thought process and to document any code that's hard to understand.

Submission

- When you've completed the assigned case study, submit its Notebook file.

The Airports case study for section 1

Prerequisite chapters: 2-4

Data file

`tsa_claims1.csv`

Scenario

You are an analyst employed by the U.S. Transportation Security Administration (TSA). Your boss has asked you to produce a report that answers the following questions about insurance claims against airports in the U.S.

Questions

- What is the most common type of insurance claim?
- Which claim site within the airport are claims most commonly filed for?
- What type of claim is made most at each claim site?
- What is the typical claim amount?
- What is the overall claim approval rate for the entire U.S.?

Instructions

1. Create a Notebook for the study.
2. Clean the data and explain why you cleaned it the way you did.
3. Create a heading for the first question.
4. If necessary, perform additional cleaning or preparation for the question.
5. Create a table or plot that answers the question and use a heading or raw text to summarize what the table or plot tells you.
6. Repeat steps 3-5 for each question.

Note

- If you encounter warnings or errors as you work on this case study, search the Internet for a solution to the problem and implement it.

The Laptops case study for section 1

Prerequisite chapters: 2-4

Data file

`laptops.csv`

Scenario

You are an analyst employed by a computer hardware company. Your company will be entering the laptop market soon. Your boss has asked you to produce a report that answers the following questions about laptops to help your company better understand the market.

Questions

- How many different laptop brands are there?
- What are the names and prices of the most and least expensive laptops?
- How are laptop prices distributed?
- What are the min, max, and mean display sizes?
- What is the average price for each brand?

Instructions

1. Create a Notebook for the study.
2. Add a heading for the first question.
3. If necessary, perform additional cleaning or preparation for the first question.
4. Create a table or plot that answers the question and use a heading or raw text to summarize what the table or plot tells you.
5. Repeat steps 2-5 for each question.

The Medical Costs case study for section 1

Prerequisite chapters: 2-4

Data file

`insurance.csv`

Scenario

You are an analyst employed by a health care company. Your boss has asked you to produce a report that answers the following questions about medical insurance charges.

Questions

- How is age related to medical costs?
- How is number of children related to medical costs?
- How is the number of people distributed by region?
- How is the number of people distributed by age?
- How are the charges distributed?

Instructions

1. Create a Notebook for the study.
2. Create a heading for the question.
3. If necessary, perform additional cleaning or preparation for the first question.
4. Create a table or plot that answers the question and use a heading or raw text to summarize what the table or plot tells you.
5. Repeat steps 2-5 for each question.

Note

- If you encounter warnings or errors as you work on this case study, search the Internet for a solution to the problem and implement it.